

HeatStart: energy-stable diffusion can still ring

A reproducible finite-volume study of startup damping

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Abstract

Does decreasing numerical energy ensure physically plausible heat diffusion? An original insulated two-material experiment separates conservation, quadratic-energy stability and temperature positivity. A cell-centred finite-volume solver compares backward Euler, Crank–Nicolson and four backward-Euler half-steps followed by Crank–Nicolson. The undamped method produces a first-step minimum temperature excess of -0.554428 while reducing quadratic energy to 0.505335 of its initial value. Startup damping removes this undershoot in the recorded experiment, but is not asserted to guarantee a maximum principle for subsequent steps. Independent discrete references and a smooth continuum solution isolate time and space errors. All results are synthetic; no physical validation or new method is claimed.

1 Question and conservative model

Heat diffusion with abrupt initial data excites rapidly decaying spatial modes. A large implicit step may be energy stable while resolving these modes poorly. The question is whether a short dissipative startup improves the temperature field without sacrificing conservation or asymptotic time accuracy. Startup smoothing is an established approach [1, 2].

On $0 < x < 1$, consider $c(x)T_t = (k(x)T_x)_x$, with insulated ends, unit area, positive piecewise-constant conductivity k and volumetric capacity c . There are no sources, contact resistance or phase changes. For cell width h_i , define the mass and face conductance

$$M_i = c_i h_i, \quad g_{i+1/2} = \left(\frac{h_i}{2k_i} + \frac{h_{i+1}}{2k_{i+1}} \right)^{-1}. \quad (1)$$

Paired heat rates $g_{i+1/2}(T_i - T_{i+1})$ give $M\dot{\mathbf{T}} + K\mathbf{T} = 0$, where M is diagonal and K is a symmetric positive-semidefinite tridiagonal matrix with $K\mathbf{1} = 0$. Thus heat $H = \mathbf{1}^T M\mathbf{T}$ is constant, whereas the distinct quadratic functional $E = \mathbf{T}^T M\mathbf{T}/2$ decreases.

2 Time integration and what stability means

The theta method solves

$$(M + \theta\Delta t K)\mathbf{T}^{n+1} = [M - (1 - \theta)\Delta t K]\mathbf{T}^n. \quad (2)$$

Backward Euler (BE) uses $\theta = 1$ and Crank–Nicolson (CN) uses $1/2$. For $\mathbf{v} = \theta\mathbf{T}^{n+1} + (1 - \theta)\mathbf{T}^n$ and $\mathbf{d} = \mathbf{T}^{n+1} - \mathbf{T}^n$,

$$E^{n+1} - E^n = -\Delta t \mathbf{v}^T K \mathbf{v} - (\theta - \tfrac{1}{2}) \mathbf{d}^T M \mathbf{d} \leq 0 \quad (\theta \geq \tfrac{1}{2}). \quad (3)$$

This identity does not constrain each component's sign. A mode with $K\phi = \lambda M\phi$ has CN amplification $a(z) = (1 - z/2)/(1 + z/2)$, $z = \lambda\Delta t$; for $z > 2$ it alternates in sign while $|a| < 1$. BE instead has positive amplification $(1 + z)^{-1}$.

3 Implementation and pulse experiment

HeatStart implements the model in Python using SciPy’s compiled banded Cholesky factorization. It reuses one factor for each step size and theta value; factorization and each solve cost $O(N)$, while storing the full history costs $O(Nn_t)$. Inputs are copied and checked for finite, positive material properties and increasing cell edges. No course code or datasets are distributed. The algorithm is:

1. Assemble masses and conductances once from the mesh and properties.
2. Factor the CN full-step matrix and, for startup, the BE half-step matrix.
3. Replace each of the first two macro intervals by two BE half-steps; advance all remaining intervals with CN. Store macro-time states.

Pure BE and pure CN are integrated over the same macro-time grid. The startup transfer over its first two intervals is $(1 + z/2)^{-4}$ for each mode. The method restarts startup on every API call; an uninterrupted experiment therefore uses a single call.

All experiments use dimensionless quantities. The pulse case has 128 uniform cells on $[0, 1]$. The left half has $(k, c) = (1, 1)$ and the right half $(4, 2)$. Only zero-based cell 63 initially has $T = 1$; the other cells have $T = 0$. This is a finite, mesh-defined pulse adjacent to a material interface. We set $\Delta t = 20 \min_i (M_i / K_{ii}) = 3.052 \times 10^{-4}$ and take 12 macro steps. Negative excess temperature means an undershoot below the initial baseline.

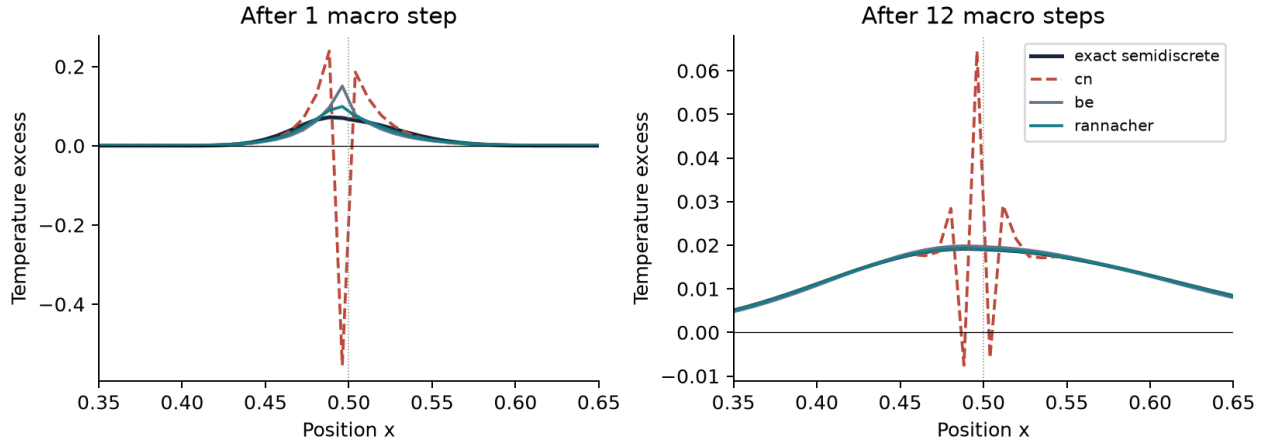


Figure 1: Original pulse experiment. The reference solves the same discrete ODE exactly in time, up to eigensolver roundoff. The dotted line marks the material interface. Axes show a neighbourhood of the pulse.

Method	First minimum	E^1/E^0	Final time error
CN	-5.544×10^{-1}	0.50534	4.747×10^{-3}
BE	1.808×10^{-13}	0.06482	2.270×10^{-4}
Startup + CN	6.271×10^{-17}	0.05435	1.231×10^{-5}

Table 1: Mass-weighted RMS error at $t = 12\Delta t$ against the discrete reference. Every table value is generated from recorded JSON outputs.

CN is energy stable but retains alternating short-scale modes (Fig. 1). Startup reduces the final error by a factor of 385.6 relative to CN at this step size. The nonnegative minimum for startup is an observation for these data, not a general theorem.

4 Verification and separation of errors

The reference diagonalizes $S = M^{-1/2}KM^{-1/2} = Q\Lambda Q^T$ and computes

$$\mathbf{T}_h(t) = M^{-1/2}Qe^{-t\Lambda}Q^TM^{1/2}\mathbf{T}_0. \quad (4)$$

Small negative eigenvalues caused by roundoff are clipped to zero. This reference requires $O(N^2)$ eigenvector storage and is used for verification, not as a large-scale solver. On a separate five-cell nonuniform heterogeneous mesh, its maximum discrepancy from a dense matrix exponential is 7.994×10^{-15} . Tests also compare individual steps with dense linear solves and the two-cell exchange mode with its closed-form amplification.

The reported norm is $\|e\|_M = (\sum_i M_i e_i^2 / \sum_i M_i)^{1/2}$. Time error compares the numerical integrator to Eq. (4) on the fixed pulse mesh at $t = 0.02$, using 20, 40, 80, 160 and 320 intervals. The last refinement pair gives orders 1.99997 (CN), 0.99917 (BE) and 2.00003 (startup). Coarse CN errors are not in the asymptotic regime; the full curves must be considered rather than a single fitted slope.

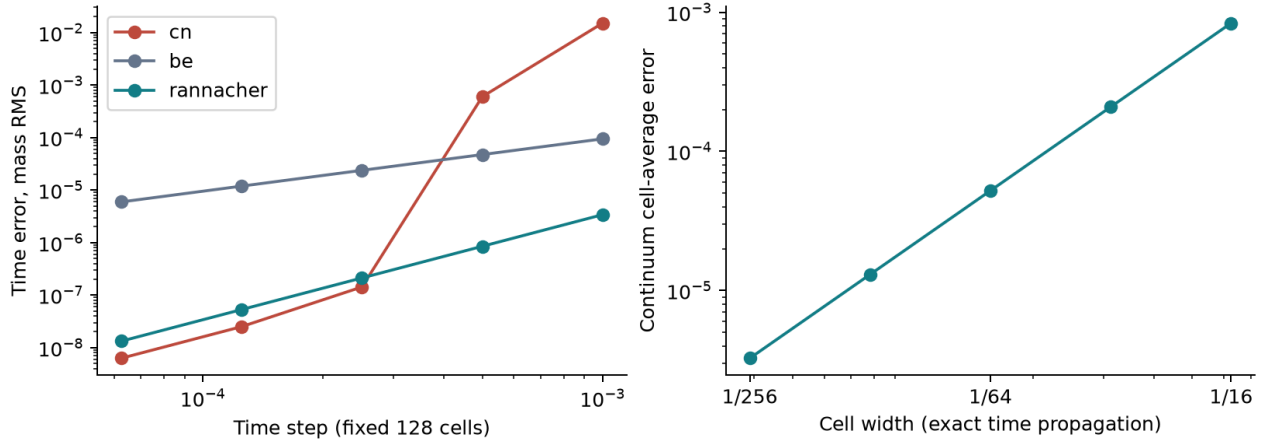


Figure 2: Left: time error at fixed spatial discretization for the heterogeneous pulse. Right: continuum cell-average error for a separate homogeneous cosine experiment with exact discrete time propagation. These are different errors.

To isolate space error, take homogeneous $k = c = 1$ and the smooth solution $T(x, t) = 2 + e^{-\pi^2 t} \cos(\pi x)$. Initialize exact cell averages $T_i(0) = 2 + \text{sinc}(h/2) \cos(\pi x_i)$, where $\text{sinc}(s) = \sin(\pi s)/(\pi s)$, and compare Eq. (4) at $t = 0.1$ with analytical cell averages. No time-stepping error is present.

Cells	Cell-average space error	Order
16	8.346×10^{-4}	–
32	2.088×10^{-4}	1.99858
64	5.222×10^{-5}	1.99965
128	1.306×10^{-5}	1.99991
256	3.264×10^{-6}	1.99998

Table 2: Spatial refinement on the smooth homogeneous problem. Orders are $\log_2(e_h/e_{h/2})$. The norm measures cell averages, not a reconstructed field.

5 Discussion, reproducibility and limitations

All three methods preserve heat within a maximum relative drift of 6.439×10^{-15} in the pulse experiment. Their quadratic energy decreases at every recorded interval, but CN violates the lower temperature bound (Fig. 3). Conservation and dissipation are therefore necessary diagnostics that do not replace a componentwise bounds check.

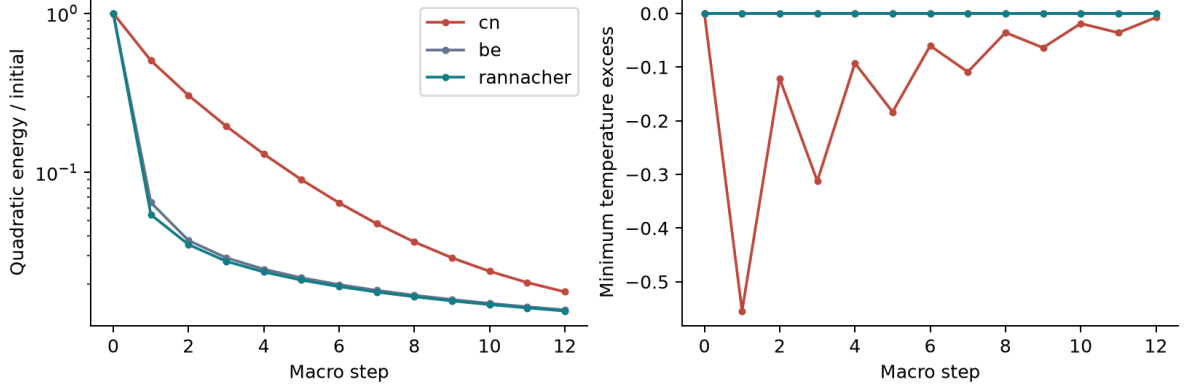


Figure 3: Conservation of heat is checked separately in the data. Here all three energy curves decrease, while only undamped CN produces a negative temperature excess in the recorded pulse trajectory.

For this discretization, $M + \theta \Delta t K$ has a nonnegative inverse. A sufficient nonnegative-update condition is $\Delta t \leq \min_i M_i / [(1 - \theta) K_{ii}]$ for $\theta < 1$. BE has no such step restriction, and CN has the sufficient bound $2 \min_i (M_i / K_{ii})$. Exceeding this bound loses a guarantee but does not prove every initial condition will undershoot. Similarly, a finite startup attenuates stiff modes but cannot certify every subsequent CN step.

CN becomes more accurate than startup on the finest tested time grids; startup is not a universal accuracy improvement. The smooth space experiment verifies only the homogeneous cell-average problem, not interface accuracy, arbitrary nonuniform meshes or physical experiments. Radiation, latent heat, convection and temperature-dependent properties are omitted.

The CLI `heatstart --output examples/results` reproduces complete CSV histories, JSON metrics and figures. Scripts check all recorded values, generate manuscript tables from JSON, compile LaTeX and enforce the page cap. Hashes bind inputs to the PDF; every rendered page is visually inspected.

6 Conclusion

The heterogeneous pulse demonstrates that CN energy decay can coexist with unphysical undershoot. Four BE half-steps reduce coarse-step time error while retaining second-order convergence. Separate references distinguish discrete verification from continuum error; physical validity remains untested.

References

- [1] R. Rannacher. Finite element solution of diffusion problems with irregular data. *Numerische Mathematik* 43, 309–327, 1984. [Public bibliographic record](#).
- [2] H. P. Langtangen. *Finite difference methods for diffusion processes*. INF5620, 2014. [Author’s public notes](#).